Name

Show appropriate work. Give real solutions. Do not solve linear systems or compute eigenvalues unless you need to.

1. Solve the ODE $2\,t\,y'(t) - 6\,y(t) = 4\,t^3$ with $y(1) = -3$

$$y' - \tfrac{3}{t}y = 2t^2$$

$$\text{I. F. } v(t) = e^{-3\ln(t)} = t^{-3}$$

$$t^{-3}y' - 3t^{-4}y = 2t^{-1}$$

$$(t^{-3}y)' = 2t^{-1}$$

$$t^{-3}y = \int 2t^{-1}\,dt + C$$

$$t^{-3}y = 2\ln(t) + C$$

$$\text{I. C. } 1^{-3}(-3) = 2\ln(1) + C$$

$$C = -3$$

$y(t) = \dfrac{2\ln(t)}{t^{-3}} + \dfrac{C}{t^{-3}}$

or $y(t) = 2t^3\ln(t) + Ct^3$

2. Complementary solutions y_c

2.1. Compute y_c for $y' = A\,y$ with A below. Describe the dominant behavior as $t \to \infty$.

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MatrixForm[Eigensystem[A = (6 -15 15; 1 2 1; 1 11 -8)]]

  -9           6           3
{-1, 0, 1}  {3, 1, 1}  {0, 1, 1}
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$$y_1 = e^{-9t}\begin{pmatrix}-1\\0\\1\end{pmatrix}$$

$$y_2 = e^{6t}\begin{pmatrix}3\\1\\1\end{pmatrix}$$

$$y_3 = e^{3t}\begin{pmatrix}0\\1\\1\end{pmatrix}$$

$y_c(t) = C_1 y_1 + C_2 y_2 + C_3 y_3$

As $t \to \infty$ $y \sim C_2 e^{6t}\begin{pmatrix}3\\1\\1\end{pmatrix}$

exp. Growing.

2.2. Compute y_c for $y' = A\,y$ with A below. Describe the dominant behavior as $t \to \infty$.

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MatrixForm[Eigensystem[A = (1 -6
                            3 -5)]]
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$$\begin{pmatrix} -2+3i & -2-3i \\ \{1+i,1\} & \{1-i,1\} \end{pmatrix}$$

$a=-2 \quad b=3$

$p = \begin{pmatrix}1\\1\end{pmatrix}$ [redacted] $q = \begin{pmatrix}1\\0\end{pmatrix}$

$Y_1 = e^{-2t}\left(\cos(bt)p - \sin(bt)q\right)$

$Y_2 = e^{-2t}\left(\sin(bt)p + \cos(bt)q\right)$

$y_c(t) = C_1 Y_1 + C_2 Y_2$

As $t \to \infty$ Decays like e^{-2t} ossc. like $\cos(3t)$

3. Compute y_c for $y' = A\,y$ with A below. Describe the dominant behavior as $t \to \infty$.

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MatrixForm[Eigensystem[A = (1 -1
                            9 -5)]]
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$$\begin{pmatrix} -2 & -2 \\ \{1,3\} & \{0,0\} \end{pmatrix}$$

$Y_1 = e^{-2t}\begin{pmatrix}1\\3\end{pmatrix}$

$Y_2 = te^{-2t}\begin{pmatrix}1\\3\end{pmatrix} + e^{-2t}u$

u satisfies

$(A+2I)u = \begin{pmatrix}1\\3\end{pmatrix}$

$$\left[\begin{array}{cc|c} 3 & -1 & 1 \\ 9 & -3 & 3 \end{array}\right] \sim \left[\begin{array}{cc|c} 3 & -1 & 1 \\ 0 & 0 & 0 \end{array}\right]$$

second comp free.
choose it zero!

$u = \begin{bmatrix} 1/3 \\ 0 \end{bmatrix}$

$y_c(t) = C_1 Y_1 + C_2 Y_2$

As $t \to \infty$

$Y_c \sim C_2$ [redacted] Y_2

slightly slower decay than e^{-2t}

also fine to say

$Y_c \sim C_2\, te^{-2t}\begin{pmatrix}1\\3\end{pmatrix}$

4. Particular solutions y_p

4.1. Write down the y_p form for $y' = A\,y + b$ with $b = \begin{pmatrix} -2 \\ 4 \end{pmatrix}$.

$y_p(t) = u$

$y_p = u \qquad 0 = Au + b$

$y_p' = 0 \qquad -Au = b$

Write down the Lin System for the coefficients

$-Au = b$

$[-A \,\vdots\, b]$ Aug mat version.

4.2. Write down the y_p form for $y' = A\,y + b$ with $b = e^{-2t}\begin{pmatrix} 3 \\ 7 \end{pmatrix}$.

$y_p(t) = u e^{-2t}$

$y_p = u e^{-2t}$

$y_p' = -2u e^{-2t}$

$-2u e^{-2t} = Au e^{-2t} + e^{-2t}\begin{pmatrix} 3 \\ 7 \end{pmatrix}$

$-2u = Au + \begin{pmatrix} 3 \\ 7 \end{pmatrix}$

$-Au - 2u = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$

$(-2 - 2I)\,u = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$

Write down the Lin System for the coefficients

$-Au - 2u = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$

Aug mat version

$\left[-A - 2I \,\middle\vdots\, \begin{matrix} 3 \\ 7 \end{matrix}\right]$

5. Particular solutions y_p

5.1. Write down the y_p form for $y' = A\,y + b$ with $b = \begin{pmatrix} 1+t^2 \\ t \\ 2 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + t\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + t^2\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

$$\boxed{y_p(t) = u_0 + t\,u_1 + t^2 u_2}$$

5.2. Write down the y_p form for $y' = A\,y + b$ with $b = \cos(4\,t)\begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

$$\boxed{y_p(t) = \cos(4t)\,u_1 + \sin(4t)\,u_2}$$

Write down the Lin System for the coefficients

$$\boxed{\left[\begin{array}{cc|c} -4I & -A & \begin{matrix}0\\0\end{matrix} \\ -A & 4I & \begin{matrix}1\\2\end{matrix} \end{array}\right]}$$

Any mat version

$y_p = \cos(4t)\,u_1 + \sin(4t)\,u_2$

$y_p' = -4\sin(4t)\,u_1 + 4\cos(4t)\,u_2$

$-4\sin(4t)\,u_1 + 4\cos(4t)\,u_2 = \sin(4t)\,A u_2 + \cos(4t)\,A u_1 + \cos(4t)\begin{pmatrix}1\\2\end{pmatrix}$

so

$-4u_1 = A u_2$

$4u_2 = A u_1 + \begin{pmatrix}1\\2\end{pmatrix}$

$-4I\,u_1 - A u_2 = \begin{pmatrix}0\\0\end{pmatrix}$

$-A u_1 + 4I\,u_2 = \begin{pmatrix}1\\2\end{pmatrix}$

$$\begin{bmatrix} -4I & -A \\ -A & 4I \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix}$$